the canonical measure, we can find many functions that have the same average. For example, Rugh [319] suggested the formula

$$\frac{\mathrm{Av}_\beta(\|\nabla U\|^2)}{\mathrm{Av}_\beta(\Delta U)} = k_\mathrm{B}\mathrm{T}. \tag{6.5}$$

We can see this as a consequence of the following proposition (see [193, 211]).

Proposition 6.1 *Let H be a Hamiltonian defined on the phase space $\mathbb{R}^{2\mathrm{N_c}}$ and suppose $\boldsymbol{G} : \mathbb{R}^{2\mathrm{N_c}} \to \mathbb{R}^{2\mathrm{N_c}}$ is a smooth (C^1) vector field with the following properties:*

- $0 < |\mathrm{Av}_\beta(\boldsymbol{G} \cdot \nabla H)| < \infty$,
- $0 < \mathrm{Av}_\beta(\nabla \cdot \boldsymbol{G}) < \infty$,
- $|\boldsymbol{G}e^{-\beta H}| < \infty$

Then

$$\mathrm{k_B T} = \frac{\mathrm{Av}_\beta(\boldsymbol{G} \cdot \nabla H)}{\mathrm{Av}_\beta(\nabla \cdot \boldsymbol{G})}.$$

Proof The proof relies on a $2\mathrm{N_c}$-dimensional version of integration by parts. Let $\mathcal{B}_R$ be the $2\mathrm{N_c}$-dimensional ball centered at the origin, of radius R, and let Γ_R be its boundary, the $2\mathrm{N_c} - 1$-dimensional sphere. Then, for any smooth scalar field g defined on $\mathcal{B}_R$, we have

$$\int_{\mathcal{B}_R} \boldsymbol{G} \cdot \nabla g \mathrm{d}\omega = \int_{\Gamma_R} (\boldsymbol{G}g) \cdot \hat{\boldsymbol{n}} \mathrm{d}\Gamma - \int_{\mathcal{B}_R} (\nabla \cdot \boldsymbol{G}) g \mathrm{d}\omega, \tag{6.6}$$

where $\mathrm{d}\omega = \mathrm{d}q_1 \ldots \mathrm{d}p_{\mathrm{N_c}}$ is the standard measure of Lebesgue integration in $\mathbb{R}^{2\mathrm{N_c}}$ and $\mathrm{d}\Gamma$ is the surface area element of the hypersphere, and $\hat{\boldsymbol{n}}$ represents the unit outward normal to the sphere. Replace g by $\rho_\beta = \exp(-\beta H)$, then it follows that

$$\left| \int_{\Gamma_R} (\boldsymbol{G}\rho_\beta) \cdot \hat{\boldsymbol{n}} \mathrm{d}\Gamma \right| \leq \max_{\Gamma_R} |\boldsymbol{G}\rho_\beta| \sigma_R,$$

where σ_R is the surface area of the hypersphere. Divide (6.6) by the volume v_R of $\mathcal{B}_R$. Then, for large R, since the ratio of surface area to volume is proportional to R^{-1}, we have

$$\frac{\int_{\mathcal{B}_R} \boldsymbol{G} \cdot \nabla \rho_\beta \mathrm{d}\omega}{v_R} \approx -\frac{\int_{\mathcal{B}_R} (\nabla \cdot \boldsymbol{G}) \rho_\beta \mathrm{d}\omega}{v_R}.$$

Since $\nabla \rho_\beta = -\beta \nabla H \rho_\beta$, we obtain, in the limit $R \to \infty$, the desired result. □